

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read: at the patch order of a simulated cortical sheet it constrains at most 10 sites, so on a larger population the positive link from coherence to content rests entirely on a bridge assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps

and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows consistency, not cortical realization. The conditions yield a disconfirmable comparison: the agreement of decoders fitted independently on two regions, in excess of what their separate accuracies imply, should be larger for perceived than for unperceived stimuli at matched phase coherence and single-region accuracy, with perception classified by a no-report marker. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. No principle yet selects the cover, and the sheaf’s overlaps are spatial where the example’s are overlaps of content, so the proposed correlate of unity has no operational definition. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction

13 from an internal encoding). The mathematical states are spatial measures
14 and fields; identifying them with perceptual content requires independent
15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which. Phase
23 order constrains shared content only through encoders related on overlaps,
24 with a bound that grows as the square root of the population it is read
25 on. Compatible local descriptions, not synchrony, determine a unique global
26 state. A contracting self-map fixes one state but can reconstruct accurately
27 only a family of bounded diameter. Each result names a measurement that
28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
30 Critiques of the temporal binding hypothesis argued that synchrony, where
31 it occurs, cannot say which features belong together or what they jointly
32 represent: that grouping has to be computed before synchrony could sig-
33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
35 and a pigeonhole count. What formalization adds to the informal argument is

36 quantitative and bookkeeping content. It gives the rate at which a coherence
37 bound on content weakens with population size and the population at which
38 it stops constraining anything, the diameter within which one contracting
39 map can reconstruct, and the number of codes a separated family of states
40 requires. It also gives a complete inventory of the premises a synchrony-to-
41 content argument consumes, each of which names a measurement, together
42 with one model in which all of them hold at once.

43 The familiar noisy mean-field oscillator supplies a stationary coherence
44 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
45 condition for a declared actuator, bounds that connect phase order to con-
46 tent only through specified encoders, and conditions under which compatible
47 descriptions glue and an accurate reconstruction fixes the glued state. A
48 Lean 4 development checks these implications and a joint witness satisfies
49 their eight connecting hypotheses in a specified model. It proves neither that
50 cortex satisfies those hypotheses nor that the glued state is experienced. The
51 latter identification motivates the framework and needs its own endpoint.

52 The empirical question is which measurements could support or discon-
53 firm the proposed connections. A phase resultant can miss a locally or-
54 dered winding field; stationary phase distributions can leave the coupling
55 mechanism unidentified; and coherent adaptation can lose alignment with
56 a designated content template. Independent decoders must identify shared
57 quantities before their agreement can be tested. We therefore distinguish sta-
58 tionary branches from finite observed onset, and compare prespecified field

59 and connectivity onset models on held-out emergence recordings. The com-
60 parison tests a specified spatial prediction, not the presence of experience.
61 Section 10.2 states the comparison that bears on conscious content directly.
62 Sections 2–5 develop the three requirements; Section 6 applies their mea-
63 surement obligations to a digital candidate. Section 7 states the resource and
64 composition premises, and Sections 8–9 examine observables and the cortical
65 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
gives their derivations and controls.

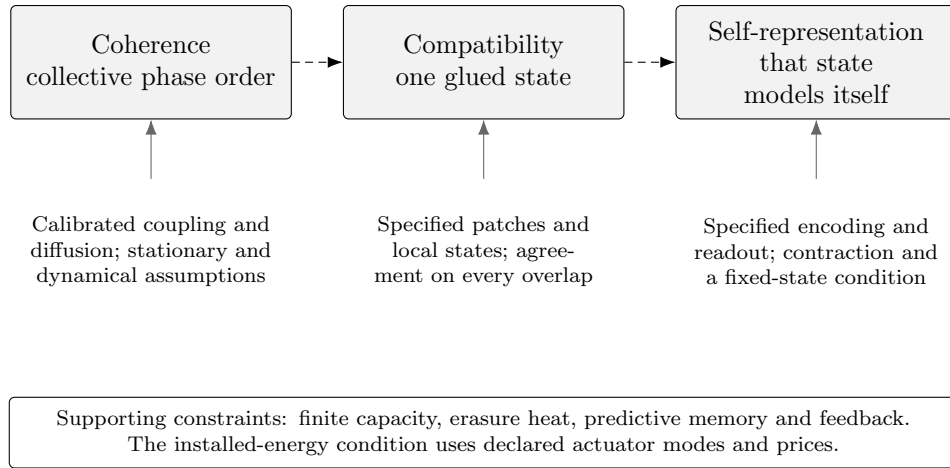


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

67 **2 Limits of phase order**

68 Several limits hold without the cortical hypothesis. Equal phases can coexist
69 with incompatible local descriptions; phase order constrains shared content
70 only through encoders related on overlaps. A winding field can have zero
71 global resultant while remaining locally ordered. A constant readout can
72 have a unique fixed point without reconstructing its input. A region that
73 does not distinguish relevant states cannot encode them, and a site cannot
74 report a change outside its causal past by a declared deadline. These lim-
75 its reflect familiar phase oscillator and information constraints (Acebrón et
76 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
77 The sections below state each one beside the condition or measurement it
78 motivates.

79 **3 Coherence: physical coupling and its re-** 80 **source condition**

81 **3.1 Dynamics and the stationary threshold**

82 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

83 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
84 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
85 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
86 quency heterogeneity and diffusion are distinct parameters. The stationary
87 calculation below uses identical frequencies and this scalar coupling. Ap-
88 plying it to a spatial kernel requires an aggregation rule and a check of the
89 resulting approximation.

90 For identical frequencies in their rotating frame, the positive normalized
91 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

92 where I_n are modified Bessel functions and ψ is the collective phase. The for-
93 mal derivation uses the absence of stationary current for a periodic gradient
94 drift to obtain this density, then verifies normalization and self-consistency.
95 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
96 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
97 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

98 Figure 2 shows this stationary branch.

99 Linearization about the uniform density gives rate $K/2 - D$ for the first
100 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists

101 exactly above threshold. About the coherent density, the quadratic form
 102 is strictly dissipative transverse to collective rotation, whose rate is zero.
 103 These linear estimates concern the limiting equation. Relating them to finite
 104 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 105 tion; exchange symmetry alone does not establish the needed independence
 106 (Sznitman, 1991).

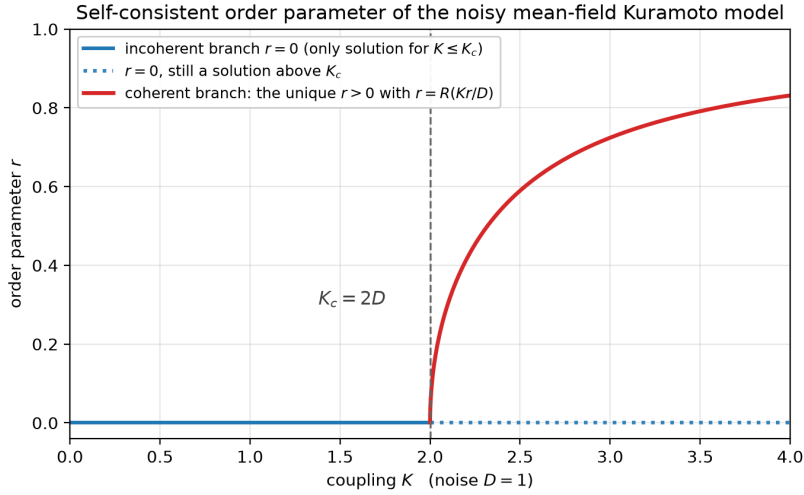


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

107 **3.2 Installed energy bounds mean coupling**

108 A spatial model replaces the finite sum by integration against a kernel. One
 109 declared implementation carries response modes with continuous profiles ϕ_i ,

110 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

111 The same configuration sets the reciprocal kernel and stored energy. An
 112 executed transition changes the kernel through its changed occupancies and
 113 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 114 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 115 pair reconstruction on refining spatial partitions converges to its product-
 116 measure integral under the specified compactness and sampling assumptions;
 117 the refinement index is independent of the number of control updates.

118 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 119 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 120 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

121 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 122 coherent stationary branch and any growing Fourier mode about the uniform
 123 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 124 is necessary: energy above it need not produce sufficient coupling because
 125 the bound can be loose.

126 The theorem concerns stored energy in the declared actuator modes. The
 127 estimated cortical signalling expenditure and extracellular electrostatic en-
 128 ergy in Supplemental Material measure different quantities; neither identifies
 129 the occupancies, mode prices or their relation to metabolic supply. Conse-
 130 quently the bound supplies a necessary condition for a declared actuator,
 131 but no cortical value of κ or demonstration that cortex meets it. Applying
 132 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 133 a justified reduction.

134 **4 Compatibility: from coordinated activity** 135 **to a global description**

136 **4.1 Why common timing is insufficient**

137 For overlapping patches U and V , compatibility means that restricting either
 138 local description to $U \cap V$ gives the same object. Descriptions can agree on
 139 shared quantities while contributing different information elsewhere. Com-
 140 patible sections of the specified sheaf have a unique global extension. On
 141 a finite cover, local finite spatial measures agreeing as measures on every
 142 overlap also have a unique finite-measure extension whose sheaf image is the
 143 glued section. No mass-one normalization is imposed.

144 In the cup example, visual and reaching descriptions must express a
 145 shared quantity, such as cup location, in common coordinates. Restriction

146 retains that quantity while dropping details particular to each description.
147 Agreement permits surface appearance and hand configuration to contribute
148 to one situation. The mathematical counterexample assigns the same phase
149 to every patch while two local mass profiles disagree on a shared site. Those
150 profiles have no common extension. Because they are chosen independently
151 of phase, synchrony places no constraint on their disagreement.

152 This distinction limits the biological interpretation. Receptive territories
153 and content domains must be related to spatial patches, and the descriptions
154 must support the required extension. Matching overlap marginals of de-
155 coded probability laws need not determine a unique global joint distribution
156 (Abramsky & Brandenburger, 2011).

157 The cover is itself a choice, and the construction constrains it in one way
158 only: synchronized patches agree where they overlap. Where they do not
159 overlap there is nothing to compare, so a cover by pairwise disjoint patches
160 satisfies that obligation for *every* assignment of local descriptions and phases
161 and the gluing theorem still returns a unique global section, which is the
162 tuple it was handed. Membership of the class then reports nothing about the
163 system beyond the patches covering it. What a gluing claim asks of a cover
164 is therefore a non-degeneracy the theorem does not assume: that distinct
165 patches meet, and that what they share is a quantity fixed independently of
166 the agreement it is being used to establish.

167 The cup example exposes a second gap, between two kinds of overlap.
168 The sheaf is built over a spatial substrate, so its patches overlap where they

169 share sites. Visual and motor territories that agree on cup location overlap
 170 in what they describe, not in where they lie, and the construction supplies
 171 no relation between the two. One route recasts the construction over content
 172 variables: each region becomes a partial decoder whose domain is the set of
 173 variables it estimates, and two regions overlap exactly where they estimate
 174 a common variable. The gluing argument needs only a cover and restriction
 175 maps, so it applies formally to such a cover, but no result here is stated
 176 for one, and which variables and decoders form it is an empirical question.
 177 Until a principle selects the cover, the proposed correlate of unity has no
 178 operational definition: gluing on a declared cover is proved, and which cover
 179 cortex realizes is open.

180 **4.2 When coherence does constrain content**

181 A declared relation between phase and content gives a positive result. Let
 182 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 183 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 184 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 185 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 186 sites; their equality on overlaps is a supplied relation between descriptions of
 187 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 188 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 189 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.

190 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (6)$$

191 Here discrepancy is the norm of the difference, and r and N are read on
192 whichever population the bound is applied to. That choice is what decides
193 whether the estimate says anything. A patch carries an order parameter
194 of its own, and reading the bound there carries that patch's resultant and
195 that patch's site count, so a large population observed through small patches
196 is constrained by the small count. Read instead on the whole array, the
197 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
198 from an average over N sites. The order cannot be improved in general:
199 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
200 observations agree exactly at either scale. A shared scalar encoder is a special
201 case. Dropping the overlap relation permits disagreement even at perfect
202 locking. A common nonexpansive linear projection preserves agreement, but
203 agreement of a projection does not establish agreement of the full vector.

204 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
205 constrains anything only where the right-hand side falls below that. At the
206 sheet's own patch order, the bound is informative for at most 10 sites; beyond
207 that, the coherence-to-content link requires the bridge assumption E78 of
208 the cortical hypothesis (Table 1 gives the full composition and its edges).
209 Supplemental Material provides the detailed derivation of this saturation,

210 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 211 by a count of disagreeing pairs, a chain across a cover's shared sites, or a
 212 spectral gap at a frequency-locked configuration.

213 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 214 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 215 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 216 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

217 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 218 duce an initial mismatch. Approximate observations leave a nondecaying
 219 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 220 update to the phase process.

221 4.3 What approximate agreement determines

222 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 223 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 224 Weights that vanish outside their patches and sum to one at each site select
 225 a global profile by weighted averaging. The selection lies within ε of each
 226 patch there, and any two such selections differ by at most ε , independently
 227 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 228 $L\varepsilon$ across selections.

229 At zero tolerance, the selection is the unique exact extension. At every

230 positive tolerance there are families with no exact extension; selection then
 231 gives an approximation whose dependence on weights is bounded. Supple-
 232 mental Material gives a three-site witness attaining the bound. The mass-
 233 profile construction uses the scalar case of the content dynamics and addi-
 234 tionally requires nonnegativity of the initial profiles and observations. The
 235 general vector result supplies no identification with the spatial-measure sheaf.

236 These constructions concern declared objects: a specified cover, contents
 237 valued in a normed space, and encoders related on overlaps. They select
 238 neither the neural cover nor the decoder, and identifying their contents with
 239 perceptual content, or establishing that the identification bears on experi-
 240 ence, is a further modelling task; Section 8 sets out what a measurement of
 241 it would have to establish.

242 **5 Self-representation: a state that models it-** 243 **self**

244 **5.1 The reflexive construction**

245 Compatibility gives a global state. Reflexivity relates it to an encoding within
 246 that state. Equip global sections with a complete metric. If the additional
 247 restriction-resonance condition identifies encoding with restriction to a region

248 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

249 The region and readout are fixed across the state space. If F contracts and
250 fixes the section s_* formed from the local descriptions, Banach's theorem
251 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
252 (Table 1 gives the full composition and its edges) supplies the contraction
253 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
254 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
255 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
256 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density
257 grows (Section 3), a rate about a different state from the one F acts on,
258 and no result derives a Lipschitz law for an encoding or readout from the
259 phase dynamics. Constructing a map with that factor remains a modelling
260 obligation. A fixed contraction can reconstruct accurately only a restricted
261 family of states: if its error is at most ε throughout that family, any two
262 members are at most $2\varepsilon/(1 - \Lambda)$ apart.

263 In the cup example, the global state includes the represented scene and the
264 perceiver's relation to it. A local encoding must retain enough information
265 for the readout to recover that modelled state; cup position alone would be
266 insufficient if hand configuration and viewpoint could not be reconstructed.

267 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-

tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. For the same fixed contracting map used by E89, however, the diameter bound above is an additional restriction: at zero error the family contains at most one metric-distinct state. A broad family therefore requires a different, input-conditioned reconstruction map, one map F_u with its own fixed point for each input u , with its own accuracy

286 and stability assumptions. No such map is formalized or constructed here.
 287 Under E89 the proposed minimal-self correlate is therefore a condition on
 288 one self-state, not a model of a self that follows a changing scene. A blind
 289 encoder or constant readout fails the stated criterion for separated states
 290 despite having a unique fixed point.

291 Where the encoding is read as the state's own restriction to a region, the
 292 criterion becomes a condition on the region. Restriction resonance requires
 293 the encoder and that restriction to identify the same pairs of states, so a
 294 boundary reconstructing two relevant states more than 2ε apart while its
 295 region reads them identically is not resonant. Read forwards, a resonant
 296 boundary that reconstructs a pairwise separated family restricts injectively
 297 on it. Which states a region reads identically is fixed by where the region
 298 sits, before an encoding is chosen, so the resolution the family demands is
 299 the region's and no readout supplies it. The implication runs one way: a
 300 region that does resolve the family is not thereby shown to carry a resonant
 301 encoding, and E89 supplies no such resolution.

302 Resonance and resolution are jointly satisfiable without a local fold. Take
 303 the region to be the whole substrate and use the identity readout: restric-
 304 tion is the identity, so this boundary is resonant and reconstructs every de-
 305 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 306 E89's strict contraction premise on a nontrivial metric space. A properness
 307 condition on the region is therefore a separate requirement, and the fixed-
 308 contraction diameter bound remains in force when E89 is also required. With

309 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 310 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 311 so the family is no larger than the δ -packing number of what the region car-
 312 ries. That is the shape a continuous family of states needs, and at the whole
 313 substrate it says nothing, because the region's range is everything. The gain
 314 L is declared and derived here for no physical readout, and at unbounded
 315 gain the bound says nothing.

316 **5.3 Agreement after a change takes communication**

317 Snapshot compatibility does not specify how quickly descriptions respond
 318 when a shared quantity changes. In a synchronous network whose updates
 319 read only the site's own state and declared neighbours' messages, two initial
 320 configurations agreeing on a site's causal past through T rounds produce the
 321 same state there at round T . The deadline consequence is the one Section 2
 322 states, and it follows from the reconstruction bound in Section 5.2: a report
 323 from that site is a readout of a code that the two interventions leave identical.

324 A three-site delay line witnesses failure after one round and exact report-
 325 ing after two; cutting the path prevents delivery at every round. A report
 326 may still happen to be correct about one fixed world. The bound concerns
 327 responsiveness across interventions.

328 Throughout this section *prediction* names a reconstruction map, and the
 329 relevant family of states, the metric, the tolerance, the communication graph
 330 and the deadline are all declared inputs, as are the represented variables and

the fidelity counted as accurate. Reading the map as temporal neural prediction would require dynamics; reading the resulting code count as physical memory capacity would require a hardware model, which a state-dependent readout would have to extend.

6 What the conditions constrain in a digital candidate

The conditions bear on a large language model (LLM) running on graphics processing units (GPUs) only through measurements, and only for a declared task. For a finite spatial coupling operator with eigenvalues $|\lambda_1| \geq \dots \geq |\lambda_N|$, any linear encoder and readout factoring through d real coordinates pays reconstruction error at least $\sum_{i>d} \lambda_i^2$. That rank premise is a property of an explicitly linear bottleneck; query/key dimension does not supply it for softmax attention. On one synthetic two-view task whose target is the glued state and a self-relative coordinate, a rank-two linear decoder has predicted training-error floor 2.000 and held-out error 2.000 on a parity-held-out family. The deadline premise is similarly specific: within one forward pass, the causal mask confines position t to its causal past, so an intervention outside it cannot be reconstructed there by that step. These results rank no architecture family, imply no general inferiority of GPU hardware, and settle no question of machine experience; a claim of unity or self-representation for such a system commits its maker to identifying the shared quantities, the code and its

352 separation of the declared family under intervention. Supplemental Material
 353 gives the bottleneck derivation, the task controls, the information ceiling of
 354 token outputs and the hardware scope.

355 7 Resource premises and composition

356 Two thermodynamic statements enter, with different standing. The first is
 357 proved for the phase model: for a smooth positive periodic density with prob-
 358 ability current J and diffusion D , the current functional $\sigma_J = \int J^2/(D\rho) d\theta$
 359 bounds how fast order can change, $\int_0^\tau \sigma_J dt \geq [\arcsin r(\tau) - \arcsin r(0)]^2/(D\tau)$
 360 while $0 < r < 1$, and stationary order has no positive floor. The second is
 361 conditional: the installed-energy condition of Section 3.2 is necessary for its
 362 declared actuator, and restricts a physical candidate only once mode prices,
 363 coupling reduction and diffusion are measured. Neither is a heat or power
 364 statement until the current is identified with a physical flux, a reservoir at a
 365 stated temperature is specified and local detailed balance holds for the steps
 366 concerned; only then does σ_J convert to entropy production and the finite
 367 register budgets to Landauer heat (Landauer, 1961; Norton, 2005; Shenker,
 368 2000). No composition edge derives coherence, compatibility or reconstruc-
 369 tion from a heat bound. Finite-register capacity obeys $H(p) \leq \log |X|$, and
 370 a separate boundary result concerns a field reaching disconnected vacuum
 371 components (Kibble, 1976); Table 1 states the connecting assumptions. Sup-
 372 plemental Material gives the current-cost derivation, predictive-memory and

373 feedback balances, learning and repeated-operation budgets, and the sensi-
 374 tivity of the metabolic comparison to its assumptions.

375 **7.1 What the composition establishes**

376 The passive and feedback composition theorems each accept eight connecting
 377 implications (Table 1). In the active branch, a named substep at the register’s
 378 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

379 with B the register’s heat. The upper allocation follows from a ledger whose
 380 operations deliver that total heat to one reservoir and whose other opera-
 381 tions do not increase joint entropy. Those ledger assumptions are physical
 382 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
 383 plies them. Finite two-bit witnesses exhibit positive forward and reverse
 384 elementary paths, local detailed balance at one thermal scale, and interac-
 385 tion energies of order $k_B T$; these conditions alone do not quantify proximity
 386 to equilibrium. Cortex has not been shown to satisfy these premises. The
 387 illustrative cortical metabolic expenditure over one region and one phase-
 388 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
 389 with the register heat budget B requires a common physical ledger. Read-
 390 ing the register at the scale where its premises are physical, such as a single
 391 channel, requires E45Active to carry that step’s budget to the convergence

392 of a continuum kernel. The active branch is an alternative route to the same
393 node rather than a stronger one, and no result of Section 3.2 passes through
394 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.

Table 1 (continued)

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

395 Both branches also take a thermodynamic cover carrying exact over-
396 lap compatibility and a reflexive boundary specifying encoding and read-
397 out. They conclude that the glued section is the unique fixed point of self-
398 prediction. Compatibility and accuracy at that section enter as premises;
399 uniqueness follows. Identifying encoding with restriction needs the separate
400 resonance condition of Eq. (8).

401 Each of the eight edges is supplied rather than derived from the one above
402 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
403 pendent roots rather than a step from one to the other; both E45 discharges
404 take their convergence from refinement data and not from the thermody-
405 namic bound; E78 states that coherence relaxes the supplied cover, and the
406 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
407 coders already shared on overlaps. The inferential content of the composition
408 is its last step, where contraction and a fixed glued section give uniqueness.
409 What composing them establishes is that the premises are consistent, named
410 in one place and machine-checked together, and not that coherence follows
411 from installed energy or self-representation from coherence.

412 Joint witnesses combine a register, boundary, refining mesh and small
413 spatial cover; active witnesses add controlled thermal updates and a priced
414 installation. They establish satisfiability under their stated identifications.
415 Mesh convergence follows from refinement data, and the witnesses establish
416 no common cortical mechanism. Supplemental Material maps the claims to
417 their Lean declarations and records the counterexamples. The development

418 declares no additional axioms beyond its libraries' foundational axioms.

419 8 What the observations establish

420 The composition holds under its premises. Whether observation can estab-
421 lish those premises is a separate question, and the three requirements it
422 connects concern different objects: phase dynamics, local descriptions and
423 internal encodings. Their observables must retain these distinctions. The
424 following controls show which mechanisms remain indistinguishable, which
425 temporal approximations fail, which states one observable cannot separate,
426 and what independent content measurements must establish. Each is a finite
427 numerical regime on a declared model, with full parameters and controls in
428 Supplemental Material, and none of them calibrates a cortical quantity.

429 8.1 Stationary distributions leave the mechanism un- 430 determined

431 Equation (2) gives a concentration-coherence relation with no fitted shape
432 parameter. Estimate r from circular resultant length and a from density
433 shape: inverting R from the measured r would make agreement automatic.
434 Even independently estimated agreement identifies a statistical family shared
435 by different mechanisms.

436 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
437 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$

438 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
439 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
440 and diffusion D_0 have that same density. Their stationary distributions co-
441 incide window by window despite different causes.

442 Resolved dynamics supply additional information. In the ideal model
443 with continuous latent paths, quadratic variation accumulates at rate $2D$
444 and conditional increment means constrain drift; a filtered observation of
445 the phase needs its own observation model before either is read off. Separat-
446 ing endogenous coupling from a matched common drive additionally requires
447 a measured drive or controlled perturbation; identifying electromagnetic cou-
448 pling requires evidence specific to that interaction. Supplemental Material
449 gives the equivalence construction and its numerical check.

450 **8.2 Finite ramps change the observable onset**

451 The square-root law in Eq. (3) describes stationary solutions. A changing
452 coupling requires time for the phase distribution to approach those solutions.
453 In a deterministic mean-field control that crosses the threshold symmetrically,
454 none of the scanned rates meets the declared tracking tolerance: slowing
455 the crossing does not remove the terminal residual under that initialization.
456 A slow control variable therefore supplies no general guarantee of branch
457 tracking at onset. Supplemental Material gives the residual, rate range and
458 admissible rates away from threshold.

459 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas

460 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 461 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 462 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 463 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 464 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 465 onset exponents differ from those of the exact stationary branch evaluated
 466 on the same grids. At the fastest speed the ensemble mean never reaches the
 467 fit window, so its fitted slope does not estimate stationary onset.

468 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 469 makes the apparent exponent depend on ramp speed and fitting window.
 470 The specified deviation-based critical ramp speed is not bracketed by this
 471 sweep. With fluctuating coupling, the relevant stationary comparison also
 472 depends on rate: the mean coupling approximates order in the tested fast
 473 regime, while averaging stationary order over coupling works for the tested
 474 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 475 neither approximation.

476 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 477 $v^{0.444}$. The interval from fitting those means is $[0.394, 0.494]$; resampling
 478 replicas independently within each speed gives a 95% interval $[0.398, 0.496]$,
 479 with fraction 0.019 of 10000 bootstrap fits at or above $1/2$. This fraction
 480 describes the resampling distribution under the fitted protocol; it is not a
 481 test of the ideal exponent. A deterministic mean-field run seeded at the

fluctuation floor gives exponent 0.425 with the same estimator and escape
 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls
 show that the criterion affects the fitted exponent, but do not prove that it
 accounts for the entire finite-population shortfall. The larger-population low-
 threshold fit has substantial log residuals; matched-speed fits do not converge
 toward one-half across the tested sizes and criteria. Some trajectories also
 cross the absolute escape level before K_c , so the first postcritical crossing can
 yield a zero or grid-limited delay. This finite sweep establishes no common
 asymptotic exponent.

The ideal delayed-onset calculation motivates a square-root scaling hy-
 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 tify the coupling medium. A within-subject emergence design can vary the
 propofol taper and test how an operational onset changes with that manipu-
 lation, but interpreting the result as a coupling-rate law requires a measured
 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 describes how D could be estimated from resolved phase traces.

Additional controls address integration bias, population size and spatial
 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2.

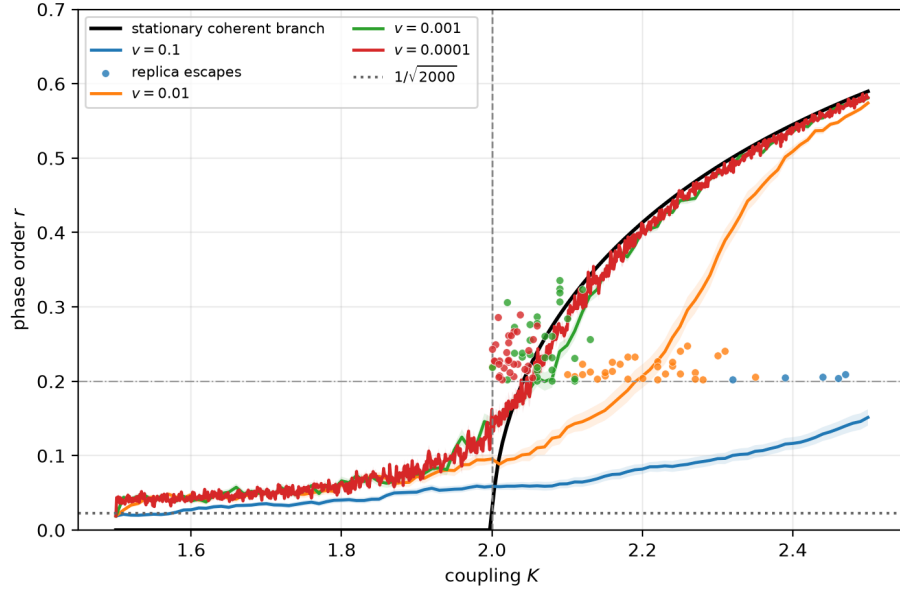


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica’s first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

502 8.3 A winding state leaves the order parameter am- 503 biguous

504 The global order parameter r discards spatial arrangement. On a periodic
505 sheet, a phase field can wind once around the sheet and have $r = 0$ while
506 neighbouring sites remain locked. At identical frequencies, every such wind-
507 ing is stationary for an isotropic symmetric kernel: contributions at opposite
508 separations cancel. It violates the quarter-turn initial-spread premise of the

509 convergence result, which is why that result can still select the uniform state.

510 In the spatial control, the winding persists below the interpolated de-
 511 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
 512 and winding starts under the same coupling, diffusion and kernel end at
 513 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 514 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 515 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 516 realizations. These are finite controls, not a cortical wave identification or a
 517 derived transition.

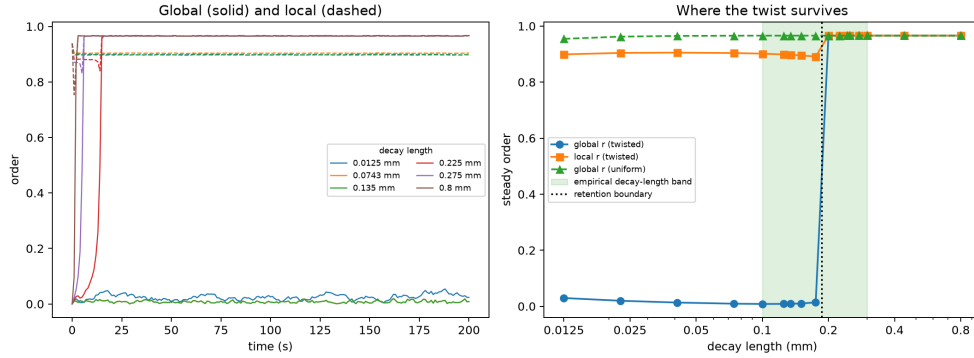


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

518 Linearization about a winding reweights each separation’s coupling by

519 the cosine of its phase advance. The resulting symmetric kernel has explicit
 520 mode rates: nonnegative reweighted coupling gives no growing mode, while
 521 a positive rate destabilizes that winding. The criterion concerns linear rates,
 522 not asymptotic stability, and the measured retention boundary need not
 523 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 524 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 525 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 526 winding from that model even though the phase-only equations retain it as
 527 a stationary state.

528 Neither model turns these states into measured cortical spirals. The pe-
 529 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 530 no propagation speed, and the amplitude variable has no calibrated relation
 531 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 532 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 533 winding patch when shared encoders are supplied, although its global $r = 0$
 534 form is uninformative. A nonzero ring winding number q also obstructs a
 535 global real-valued phase lift; it has no identified content meaning here. Thus
 536 an onset criterion based only on r can register a change of basin as well
 537 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 538 observable.

539 8.4 Coherent adaptation can lose environmental align- 540 ment

541 An adaptive coupling rule might supply representational structure, but its
542 objective must reward the relevant agreement. For symmetric coupling, pair-
543 wise sine terms cancel in the population mean drift. Minimizing squared drift
544 at that fixed mean therefore rewards equalizing drift across nodes. In the
545 frequency-cluster control, this favors coupling between unlike-frequency clus-
546 ters, whereas the diagnostic environmental template rewards coupling within
547 clusters.

548 The formalization proves descent of the squared-drift objective for a spec-
549 ified gradient flow at fixed phases. Its functional differs from thermodynamic
550 entropy production: at a coherent stationary density the probability current
551 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
552 rial gives the operator, its gradient and the conditions of the descent result.

553 The finite control co-evolves noisy phases and symmetric, nonnegative
554 couplings at fixed total resource. Its template is a diagnostic covariance
555 of latent signals, never a training input. Across three illustrative seeds,
556 second-half order remains between 0.9167 and 0.9222, while the within-
557 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
558 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
559 trols are reported in Supplemental Material.

560 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

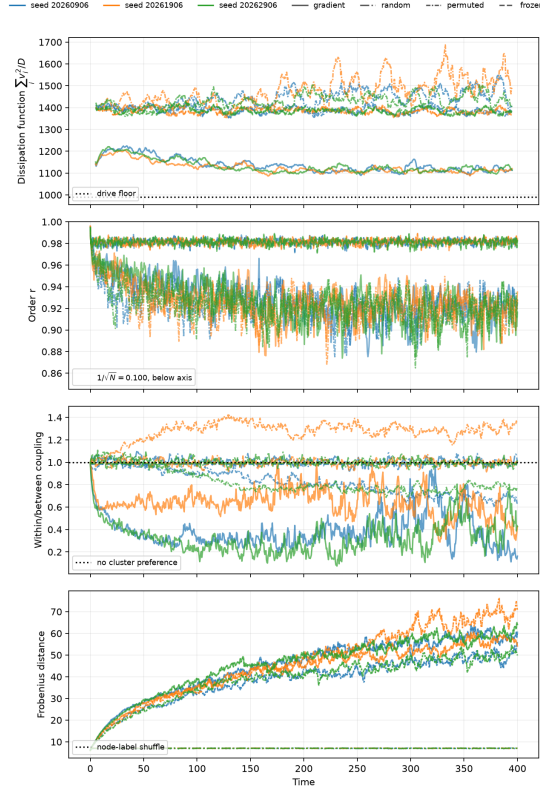


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

583 tent, phase carries a share 1.000 of the content entropy when content is built
584 from it and 0.000 when it is not. A phase-derived statistic reports com-
585 patibility on phase-locked patches whose local descriptions disagree. The
586 implemented content statistic detects their incompatibility. These controls
587 validate a measurement on known constructions; a neural content model and
588 decoder remain to be established.

589 Choosing the cover also chooses a candidate subject: trivial descriptions
590 can agree while carrying little content, and a partition can conceal disagree-
591 ment. The useful choice must therefore be constrained by what neural ac-
592 tivity encodes. The exact composition assumes compatibility of its supplied
593 cover. The content mechanisms above identify additional hypotheses under
594 which agreement can arise; they leave the empirical choice of content and
595 the connection to that cover explicit.

596 **8.6 Estimator calibration on exploratory EEG**

597 The concentration range and the dependence structure decide whether the
598 concentration-coherence relation can be tested at all. Under the specified
599 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
600 requires 3000 independent sites in the scanned designs, rising to 31000 when
601 sites share a mean in groups of 10. These counts describe that estimator and
602 dependence model at a declared design concentration; the intended recording
603 requires its own calibration. Supplemental Material reports the bin-count
604 controls and the full dependence-aware design.

605 The estimators were also exercised on public propofol-sedation EEG (Ba-
 606 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
 607 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
 608 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
 609 the residual from the Bessel relation is small. That agreement is uninforma-
 610 tive. Pooling over a window does not remove the common rotation of the
 611 phases, so a perfectly phase-locked 10-Hz signal passed through the same
 612 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
 613 instantaneous spatial resultant is one. The bipolar montage also removes
 614 spatial coherence: a field whose raw channels have instantaneous resultant
 615 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform
 616 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
 617 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
 618 the estimator’s behaviour and establishes no spatial concentration, coher-
 619 ence level or test of the stationary density. Sedation recordings also leave the
 620 awakening hypothesis untested.

621 An informative dataset therefore needs an estimand that removes the
 622 common rotation, a montage whose effect on spatial phase is modelled, suf-
 623 ficient concentration range or precision, calibrated estimators and an ob-
 624 servation model for dependence and source mixing. Intracranial recordings
 625 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
 626 informative concentration range, but that range must be measured. The
 627 design counts above are calculations for the specified estimator, not sam-

628 ple requirements transferable to another recording. Public mouse calcium
629 imaging provides thousands of spatially located neurons across waking, sleep
630 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
631 phase-estimation band used here. Site count, state coverage and bandwidth
632 are separate requirements.

633 **9 The cortical hypothesis and an awakening**

634 **test**

635 The cortical proposal identifies a physical source of coupling and asks whether
636 its changes help explain recovery. The tests in Section 8 determine what
637 observations must distinguish before a phase trace can support this identifi-
638 cation.

639 **9.1 Physical identification and calibration**

640 The cortical hypothesis identifies part of effective coupling with endogenous
641 extracellular electromagnetic interactions. Fields can modulate spike timing
642 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
643 propose coordination of distributed neural information (Pinotsis & Miller,
644 2022; McFadden, 2020). Phase organization under propofol motivates study-
645 ing this regime (Bastos et al., 2021), but does not measure coupling.

646 The formal field predicate requires a jointly continuous kernel, a normal-

647 ized substrate, coupling equal to the kernel average, and no single site pro-
648 viding more than half that strength. Toy constructions satisfy it. A cortical
649 realization must specify the scale at which the kernel and its reduction ap-
650 ply: calcium imaging reports different state-dependent modularity patterns
651 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

652 Cortical columns are the obvious candidate unit, and they answer one
653 of the two demands the formalism makes. As the oscillator index they are
654 a serious proposal: they are delineated by response properties and anatomy
655 rather than by the agreement they would be used to establish (Mountcastle,
656 1997), and a phase can be estimated from a column-sized territory at the
657 electrode spacings used for cortical wave measurements (Townsend et al.,
658 2015). As the content cover they are a poor one, for three separate reasons.
659 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
660 the compatibility obligation without constraining anything (Section 4). The
661 cover condition additionally requires the patches to exhaust the substrate,
662 while much of association cortex has no accepted columnar parcellation. And
663 the column’s standing as a canonical cortical unit is contested (Horton &
664 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
665 supplies the oscillator index; the content cover remains an open empirical
666 question, which the bridge assumption E78 records.

667 The spatial reduction is also testable. A row-sum aggregation at zero
668 diagonal tracks the uniform-coupling control over the tested spatial decay
669 range, while frequency heterogeneity displaces the operational threshold. The

670 fit window biases both estimates relative to $2D$; agreement with the control
671 is therefore the supported conclusion. Supplemental Material reports the
672 reduction, its numerical error and the distinct assumptions needed for the
673 kernel-average identification in Eq. (5).

674 The field commitment requires physical evidence beyond a choice of math-
675 ematical representation, and the available support comparisons do not supply
676 it. A finite matrix spread over positive-mass cells preserves its total cou-
677 pling and phase correlation exactly, while point-supported vanishing under
678 an atomless measure and a disadvantage from excluding a better-correlated
679 pair concern specified support constraints; Section 6 reads the same compar-
680 isons against a digital candidate. Neither supplies energy prices.

681 A spike-time shift alone supplies no coupling rate. The dimensionless
682 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
683 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (12)$$

684 This rate describes how induced timing shifts accumulate into drift. Nei-
685 ther it nor the installation factor κ follows from a heat bound. Testing the
686 physical threshold requires independent estimates of the interaction kernel,
687 its aggregation and diffusion in the same cortical regime.

688 9.2 Extracellular geometry and recovery

689 The proposed control variable is extracellular geometry. Extracellular vol-
690 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
691 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
692 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
693 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
694 an effective-medium description predicts that a larger extracellular fraction
695 raises conductivity and reduces field amplitude.

696 The coupling-gated hypothesis consequently predicts that extracellular
697 contraction on awakening raises E and, holding the other factors in Eq. (12)
698 fixed, raises effective coupling. This argument supplies a direction; the
699 geometry–coupling relation, its magnitude and its role in human recovery
700 require measurement.

701 For a smooth crossing at nonzero speed, evaluating the stationary branch
702 along the changing control gives a deterministic tracking approximation that
703 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (13)$$

704 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
705 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
706 proximation under the tested initialization. A separation of geometry and
707 phase time scales can support tracking away from threshold. The awakening

708 hypothesis needs a calibrated dynamical prediction for an operational onset
709 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
710 recovery latency.

711 **9.3 A protocol for the awakening experiment**

712 Neural inertia describes a behavioural barrier and hysteresis between anaes-
713 thetic induction and emergence (Friedman et al., 2010); recovery can pass
714 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
715 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
716 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
717 mechanism from a synaptic one. A possible discriminator is spatial onset
718 relative to independently estimated local field coupling and network connec-
719 tivity. The present mean-field model has no spatially varying κ and derives
720 no nucleation order, so a field-based onset map needs a separate model spec-
721 ified before the observations are inspected.

722 For that comparison, partition the recording into prespecified regions i
723 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
724 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
725 and a connectivity score c_i from a separately estimated structural network.
726 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
727 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

728 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (14)$$

729 Predicted onset is the first crossing of a region-specific threshold fixed from
730 pre-onset fluctuations. Fit the two scales on training sessions, then com-
731 pare held-out onset times and site order under the same observation model.
732 Larger field scores predict earlier onset only when baseline order and diffu-
733 sion do not reverse the ranking. This declared approximation omits interre-
734 gional coupling and saturation; a successful ranking would support its field
735 covariate, not prove ephaptic causation. Supplemental Material specifies the
736 controls and failure conditions.

737 A within-subject protocol can manipulate emergence rate, with spatially
738 resolved recordings to compare the proposed field and connectivity models:

- 739 1. **Manipulate emergence rate.** Vary the propofol infusion taper
740 across sessions within the same subject, recording drug concentrations
741 or a validated pharmacokinetic proxy to quantify the manipulation.
- 742 2. **Measure behavioural recovery.** Record time-to-response to a stan-
743 dard stimulus as the independent behavioural endpoint, rather than
744 estimating ΔK directly.
- 745 3. **Resolve spatial onset.** Identify anatomical locations where coher-
746 ence first crosses the baseline-scaled threshold and score the two maps

747 of Eq. (14) on held-out sessions. The two covariate maps are fixed
748 before the outcome sessions are inspected.

749 **4. Match physical controls.** Confirm that the observed coherence-to-
750 amplitude relation holds under the specific local diffusion D and ob-
751 servation noise. The model’s quadratic-variation rate $2D$ applies to
752 the latent phase; a band-limited analytic phase is smooth below its
753 correlation time, and on a constructed field its squared-increment rate
754 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
755 fore requires an explicit observation model for filtering and sampling.

756 Predeclare the recording window, regional fluctuation thresholds, and a
757 shared timing-error model. Fit the two scales and that observation model
758 on training sessions only; on held-out sessions, score the observed site-onset
759 times and sites that never cross by the same regional predictive rule. Report
760 site order as a secondary outcome and behavioural time-to-response sepa-
761 rately. Sessions without a crossing are censored observations, not zero delays
762 or exclusions. Compare the two scores only where independently estimated
763 field and connectivity maps predict distinguishable onset patterns at the
764 recording resolution. A held-out field-model advantage would support that
765 specified spatial predictor; it would not identify an ephaptic cause, shared
766 content or experience. Supplemental Material gives the controls and pilot
767 quantities needed to judge whether the comparison is informative.

768 10 Discussion

769 10.1 Unity and a minimal self as interpretive motiva- 770 tion

771 We propose the state glued from compatible local descriptions as a formal
772 correlate of conscious unity, and its accurate internal reconstruction as a cor-
773 relate of a minimal self. In the cup example, this would mean one description
774 that preserves the cup’s relation to the perceiver as well as the scene. Com-
775 patibility permits distinct contents and uncertainty; the shared quantities
776 must be chosen from independent evidence about what each region encodes.
777 The reconstruction need not be a verbal or autobiographical account.

778 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
779 ing the subject of an experience, present before any narrative or reflective
780 self-description (Gallagher, 2000). Two of its features motivate the proposed
781 correlate. It is not a separate object of experience but a perspective built
782 into the experienced scene; correspondingly, the reconstruction is not a sec-
783 ond representation beside the glued state but a map whose fixed point is
784 that state, read through a part of it. It is immediate rather than inferred
785 from a higher-order state; correspondingly, the fixed point needs no further
786 state that represents the first. The correspondence is structural and does not
787 show that a fixed point is experienced as a perspective. Self-model theory
788 locates the phenomenal self in a transparent model a system has of itself
789 (Metzinger, 2003), and interoceptive-inference accounts ground it in predic-

tion of the body’s own states (Seth, 2013). The present condition is narrower than either: it asks only that a declared local encoding recover the global state, including perceiver-relative variables, at a declared fidelity, and it says nothing about transparency or about which bodily variables must be modelled. Either account could supply the family of states and the fidelity that the condition leaves as inputs. Under E89 that family is one state at zero error and a family of diameter at most $2\varepsilon/(1-\Lambda)$ within error ε (Section 5.2), so the condition as proved constrains a single self-state; a self that persists through change needs the input-conditioned map that is not constructed here. Its proposed relation to experienced perspective remains an interpretation. Neither identification is among the eight hypotheses in Table 1, and satisfying them would not measure experience. The unity proposal also lacks an operational definition. The gluing theorem holds for whatever cover it is given, a disjoint cover satisfies it trivially, and the cover of interest overlaps in content rather than in space (Section 4); until a principle fixes that cover, the correlate is a schema awaiting one.

10.2 A comparison that could disconfirm the link to content

The framework predicts that conscious content tracks compatibility, not coherence. The comparison must also separate compatibility from how well each region encodes the stimulus, because two accurate decoders agree auto-

811 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
 812 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (15)$$

813 Consciously perceived stimuli are usually decoded better, so higher agreement
 814 on those trials would follow from single-region fidelity alone. At matched
 815 single-region error, agreement differs only through the covariance of the two
 816 errors, which is the term compatibility bears on: two regions describing one
 817 content should err together.

818 Present matched stimuli that are perceived on some trials and not on
 819 others, for example at masking or rivalry threshold, and fit decoders for
 820 one shared stimulus quantity independently on two regions that both encode
 821 it. Classify trials by perceptual state with a no-report measure, such as
 822 reflexive eye movements or pupil responses that track the dominant percept
 823 in rivalry (Tsuchiya et al., 2015), so that the classification does not also
 824 select for the neural correlates of reporting; report-based classification is a
 825 secondary analysis. Match the two classes on phase coherence between the
 826 regions and on each region’s own decoding error (or, where the stimulus
 827 value is uncertain, on each region’s signal strength). The statistic is the
 828 excess agreement over a fidelity-matched null: pair region A ’s decode on one
 829 trial with region B ’s decode on another trial of the same stimulus and error
 830 stratum, which preserves both marginal accuracies and removes any coupling

831 of the errors.

832 The proposed link predicts a larger excess on perceived trials at matched
833 coherence and matched single-region accuracy (Fig. 6). Equal excess across
834 classes, or excess that follows coherence and not perceptual state, counts
835 against it. A positive result is not yet specific: noise shared upstream of both
836 regions also correlates their errors, so the excess should be compared against
837 an upstream region’s decoding error, matched in the same way. A self-model
838 comparison has the same form: at matched accuracy for the scene variable,
839 the reconstruction of perceiver-relative variables, such as hand position rel-
840 ative to the target, from a declared local encoding should be more accurate
841 on perceived trials. These are correlational tests of proposed correlates, and
842 the no-report classification is itself an indirect marker of experience.

843 **10.3 What a test of the full proposal must establish**

844 A test needs separate estimates of effective coupling and diffusion, shared
845 content on overlapping regions, and reconstruction from a specified encoding
846 at a declared fidelity. The cover, relevant state family and decoder must
847 be fixed without selecting them for agreement. An independent experiential
848 endpoint is needed to evaluate the interpretation; absence of a report alone
849 does not define absence of experience. A unified episode with incompati-
850 ble decoded descriptions would challenge the unity identification. A unified
851 episode lacking the proposed self-encoding would challenge the minimal-self
852 claim. Conversely, satisfying the mathematical conditions without such an

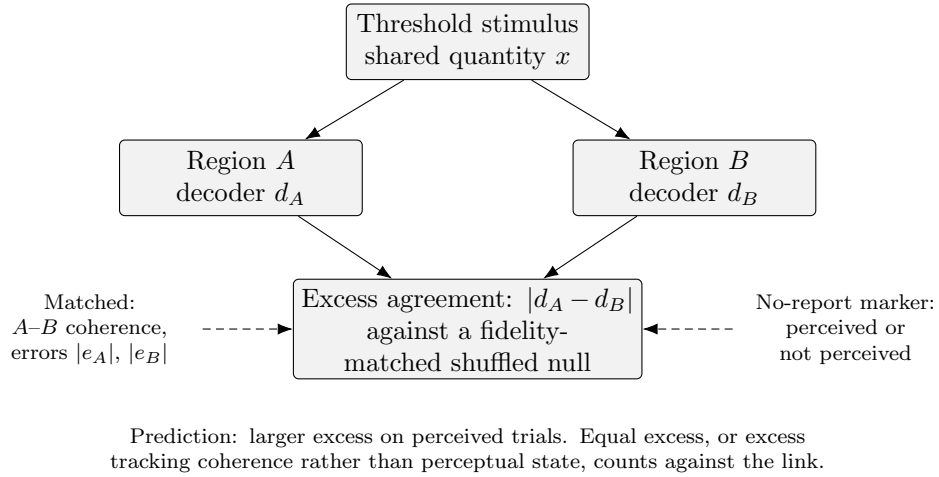


Figure 6: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (15)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

853 episode would challenge their sufficiency as correlates.

854 For the physical route, cortical actuator modes and prices, the geometry–
855 coupling relation and a dynamical onset model remain to be measured. The
856 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
857 diction. Controlled changes in coordination and independently decoded con-
858 tent agreement would address the representational connection. The synthetic
859 decoder exercise calibrates part of these measurements, and the EEG exercise
860 identifies what a pooled bipolar estimand cannot measure; neither identifies
861 a neural content code or tests experience.

862 10.4 Explanatory scope

863 Integrated-information, global-workspace and predictive-processing accounts
864 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
865 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
866 relations among coupling, local content and reconstruction. Its formal field
867 condition requires a continuous kernel, a normalized substrate, a specified
868 average coupling and noise, and a site-contribution bound. Those conditions
869 do not identify an electromagnetic substrate; the extracellular interpretation
870 needs the separate calibration of Section 9. The joint witness shows that
871 the formal hypotheses can hold together in a declared model. A cortical
872 realization and an experiential interpretation require additional evidence.

873 Four limitations bound every claim above. Synchrony is not itself an
874 experiential measure, and nothing here establishes a consciousness criterion:
875 the conditions are proposed correlates to be tested against report or another
876 independent endpoint. The present EEG analysis checks a measurement
877 pipeline and does not validate any experiential claim. The formal results are
878 conditional on declared encoders, covers and maps whose cortical counter-
879 parts are unmeasured. No principle selects the cover on which compatibility
880 is assessed, so the unity correlate is not yet operationally defined.

881 The remaining explanatory question is whether these relations occur in
882 neural activity and matter for experience. The framework identifies candidate
883 correlates and tests of their physical and representational premises. It does
884 not explain why satisfying them would be accompanied by experience.

885 11 Conclusions

886 Phase coherence, compatibility and self-reconstruction are separable condi-
887 tions. Coherence constrains shared content only through encoders shared on
888 overlaps and with a bound that grows as the square root of the population
889 read. That bound is vacuous beyond 10 sites at the simulated sheet’s patch
890 order, so for populations of cortical size the positive link from coherence
891 to content is the bridge assumption E78 and not a derived result. Com-
892 patible local descriptions, not synchrony, determine one global state; and a
893 contracting self-map fixes that state while reconstructing only a family of
894 bounded diameter. Each result replaces an assumption about synchrony by
895 a measurement: independently decoded agreement on overlaps, a declared
896 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
897 ison of Section 10.2 states how such measurements could count against the
898 proposed link to conscious content.

899 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on

Table 2 (continued)

Symbol	Defined	Meaning
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§7, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (11)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (12)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity

Table 2 (continued)

Symbol	Defined	Meaning
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (12)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Mathlib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,

908 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
909 and selection are described in Supplemental Material and implemented in
910 `simulations/empirical_collapse.py`. The repository contains the repro-
911 ducible analysis and does not redistribute the raw recordings.

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915 **Use of AI tools**

916 AI assistance to this work was substantial and is not separable by task.
917 Several coding assistants and chat interfaces were used in an interleaved way
918 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
919 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
920 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
921 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
922 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
923 they contributed to the Lean development, the simulation and analysis code,
924 and the drafting and revision of both documents. Because the tools were
925 used in combination and in alternation, no section, proof, figure or result is
926 attributable to any one of them, and none is claimed to be. The author is

927 responsible for all content, including everything a tool produced. Machine
928 checking establishes the stated Lean results under their hypotheses; it does
929 not validate biological interpretations or replace scientific review of generated
930 code and prose.

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